

Analytic Functions

Complex analysis is concerned with the theory and application of “analytic functions,” that is, functions that are differentiable in some domain, so that we can do “calculus in complex.” The definition is as follows.

DEFINITION

Analyticity

A function $f(z)$ is said to be *analytic in a domain D* if $f(z)$ is defined and differentiable at all points of D . The function $f(z)$ is said to be *analytic at a point $z = z_0$ in D* if $f(z)$ is analytic in a neighborhood of z_0 .

Also, by an **analytic function** we mean a function that is analytic in *some* domain.

Hence analyticity of $f(z)$ at z_0 means that $f(z)$ has a derivative at every point in some neighborhood of z_0 (including z_0 itself since, by definition, z_0 is a point of all its neighborhoods). This concept is *motivated* by the fact that it is of no practical interest if a function is differentiable merely at a single point z_0 but not throughout some neighborhood of z_0 . Team Project 24 gives an example.

A more modern term for *analytic in D* is *holomorphic in D* .

EXAMPLE 5 Polynomials, Rational Functions

The nonnegative integer powers $1, z, z^2, \dots$ are analytic in the entire complex plane, and so are **polynomials**, that is, functions of the form

$$f(z) = c_0 + c_1z + c_2z^2 + \dots + c_nz^n$$

where $c_0, \dots, c_n$ are complex constants.

The quotient of two polynomials $g(z)$ and $h(z)$,

$$f(z) = \frac{g(z)}{h(z)},$$

is called a **rational function**. This f is analytic except at the points where $h(z) = 0$; here we assume that common factors of g and h have been canceled.

Many further analytic functions will be considered in the next sections and chapters. ■

13.4 Cauchy–Riemann Equations. Laplace’s Equation

As we saw in the last section, to do complex analysis (i.e., “calculus in the complex”) on any complex function, we require that function to be *analytic on some domain* that is differentiable in that domain.

The Cauchy–Riemann equations are the most important equations in this chapter and one of the pillars on which complex analysis rests. They provide a criterion (a test) for the analyticity of a complex function

$$w = f(z) = u(x, y) + iv(x, y).$$

Roughly, f is analytic in a domain D if and only if the first partial derivatives of u and v satisfy the two **Cauchy–Riemann equations**⁴

$$(1) \quad u_x = v_y, \quad u_y = -v_x$$

everywhere in D ; here $u_x = \partial u / \partial x$ and $u_y = \partial u / \partial y$ (and similarly for v) are the usual notations for partial derivatives. The precise formulation of this statement is given in Theorems 1 and 2.

Example: $f(z) = z^2 = x^2 - y^2 + 2ixy$ is analytic for all z (see Example 3 in Sec. 13.3), and $u = x^2 - y^2$ and $v = 2xy$ satisfy (1), namely, $u_x = 2x = v_y$ as well as $u_y = -2y = -v_x$. More examples will follow.

THEOREM 1

Cauchy–Riemann Equations

Let $f(z) = u(x, y) + iv(x, y)$ be defined and continuous in some neighborhood of a point $z = x + iy$ and differentiable at z itself. Then, at that point, the first-order partial derivatives of u and v exist and satisfy the Cauchy–Riemann equations (1).

Hence, if $f(z)$ is analytic in a domain D , those partial derivatives exist and satisfy (1) at all points of D .

PROOF By assumption, the derivative $f'(z)$ at z exists. It is given by

$$(2) \quad f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}.$$

The idea of the proof is very simple. By the definition of a limit in complex (Sec. 13.3), we can let Δz approach zero along any path in a neighborhood of z . Thus we may choose the two paths I and II in Fig. 335 and equate the results. By comparing the real parts we shall obtain the first Cauchy–Riemann equation and by comparing the imaginary parts the second. The technical details are as follows.

We write $\Delta z = \Delta x + i \Delta y$. Then $z + \Delta z = x + \Delta x + i(y + \Delta y)$, and in terms of u and v the derivative in (2) becomes

$$(3) \quad f'(z) = \lim_{\Delta z \rightarrow 0} \frac{[u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y)] - [u(x, y) + iv(x, y)]}{\Delta x + i \Delta y}.$$

We first choose path I in Fig. 335. Thus we let $\Delta y \rightarrow 0$ first and then $\Delta x \rightarrow 0$. After Δy is zero, $\Delta z = \Delta x$. Then (3) becomes, if we first write the two u -terms and then the two v -terms,

$$f'(z) = \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x}.$$

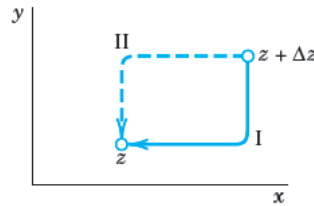


Fig. 335. Paths in (2)

Since $f'(z)$ exists, the two real limits on the right exist. By definition, they are the partial derivatives of u and v with respect to x . Hence the derivative $f'(z)$ of $f(z)$ can be written

(4)

$$f'(z) = u_x + iv_x.$$

Similarly, if we choose path II in Fig. 335, we let $\Delta x \rightarrow 0$ first and then $\Delta y \rightarrow 0$. After Δx is zero, $\Delta z = i \Delta y$, so that from (3) we now obtain

$$f'(z) = \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y)}{i \Delta y} + i \lim_{\Delta y \rightarrow 0} \frac{v(x, y + \Delta y) - v(x, y)}{i \Delta y}.$$

Since $f'(z)$ exists, the limits on the right exist and give the partial derivatives of u and v with respect to y ; noting that $1/i = -i$, we thus obtain

(5)

$$f'(z) = -iu_y + v_y.$$

The existence of the derivative $f'(z)$ thus implies the existence of the four partial derivatives in (4) and (5). By equating the real parts u_x and v_y in (4) and (5) we obtain the first

Cauchy–Riemann equation (1). Equating the imaginary parts gives the other. This proves the first statement of the theorem and implies the second because of the definition of analyticity. ■

Formulas (4) and (5) are also quite practical for calculating derivatives $f'(z)$, as we shall see.

EXAMPLE 1 Cauchy–Riemann Equations

$f(z) = z^2$ is analytic for all z . It follows that the Cauchy–Riemann equations must be satisfied (as we have verified above).

For $f(z) = \bar{z} = x - iy$ we have $u = x$, $v = -y$ and see that the second Cauchy–Riemann equation is satisfied, $u_y = -v_x = 0$, but the first is not: $u_x = 1 \neq v_y = -1$. We conclude that $f(z) = \bar{z}$ is not analytic, confirming Example 4 of Sec. 13.3. Note the savings in calculation! ■

The Cauchy–Riemann equations are fundamental because they are not only necessary but also sufficient for a function to be analytic. More precisely, the following theorem holds.

THEOREM 2

Cauchy–Riemann Equations

*If two real-valued continuous functions $u(x, y)$ and $v(x, y)$ of two real variables x and y have **continuous** first partial derivatives that satisfy the Cauchy–Riemann equations in some domain D , then the complex function $f(z) = u(x, y) + iv(x, y)$ is analytic in D .*

The proof is more involved than that of Theorem 1 and we leave it optional (see App. 4).

Theorems 1 and 2 are of great practical importance, since, by using the Cauchy–Riemann equations, we can now easily find out whether or not a given complex function is analytic.

EXAMPLE 2 Cauchy–Riemann Equations. Exponential Function

Is $f(z) = u(x, y) + iv(x, y) = e^x(\cos y + i \sin y)$ analytic?

Solution. We have $u = e^x \cos y$, $v = e^x \sin y$ and by differentiation

$$\begin{aligned}u_x &= e^x \cos y, & v_y &= e^x \cos y \\u_y &= -e^x \sin y, & v_x &= e^x \sin y.\end{aligned}$$

We see that the Cauchy–Riemann equations are satisfied and conclude that $f(z)$ is analytic for all z . ($f(z)$ will be the complex analog of e^x known from calculus.) ■

Examples of Analytic Functions

- (i) $f(z) = z$ is analytic in the whole of $\mathbb{C}$. Here $u = x$, $v = y$, and the Cauchy–Riemann equations are satisfied ($1 = 1$; $0 = 0$).
- (ii) $f(z) = z^n$ (n a positive integer) is analytic in $\mathbb{C}$. Here we write $z = r(\cos \theta + i \sin \theta)$ and by de Moivre’s theorem, $z^n = r^n(\cos n\theta + i \sin n\theta)$. Hence $u = r^n \cos n\theta$ and $v = r^n \sin n\theta$: we can check the Cauchy–Riemann equations (using $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}(y/x)$). The derivative is nz^{n-1} , as we might expect!
- (iii) $f(z) = e^z = e^x e^{iy} = e^x(\cos y + i \sin y)$. So

$$\frac{\partial u}{\partial x} = e^x \cos y = \frac{\partial}{\partial y}(e^x \sin y) = \frac{\partial v}{\partial y}; \quad \frac{\partial u}{\partial y} = -e^x \sin y = -\frac{\partial v}{\partial x}.$$

The derivative is

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = e^x \cos y + i e^x \sin y = e^z,$$

again as expected.

- (iv) $f(z) = 1/z$: check that this is analytic with derivative $-1/z^2$ in any region R which does *not* include the origin.
- (v) Any rational function – i.e., $f(z) = P(z)/Q(z)$ where P and Q are polynomials – is analytic *except* at points where $Q(z) = 0$. For instance, $f(z) = (z - i)/(z + i)$ is analytic except at $z = -i$.
- (vi) Many standard functions obey the usual rules for their derivatives; e.g.,

$$\begin{aligned}\frac{d}{dz} \sin z &= \cos z, & \frac{d}{dz} \sinh z &= \cosh z, \\ \frac{d}{dz} \cos z &= -\sin z, & \frac{d}{dz} \cosh z &= \sinh z, \\ \frac{d}{dz} \log z &= \frac{1}{z} \quad (\text{when } \log z \text{ is defined as later}).\end{aligned}$$

Examples of Non-Analytic Functions

- (i) $f(z) = \operatorname{Re}(z)$. Here $u = x$, $v = 0$, but $1 \neq 0$. $\operatorname{Re}(z)$ is *nowhere* analytic.
- (ii) $f(z) = |z|$; here $u = \sqrt{x^2 + y^2}$, $v = 0$.
- (iii) $f(z) = \bar{z} = x - iy$ (complex conjugate, also denoted z^*). Here $u = x$, $v = -y$, so $\partial u / \partial x = 1 \neq -1 = \partial v / \partial y$.
- (iv) $f(z) = |z|^2 = x^2 + y^2$. The Cauchy–Riemann equations are only satisfied at the origin, so f is only differentiable at $z = 0$. However, it is not analytic there because there is no small region containing the origin within which f is differentiable.

Example 3

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An analytic fcn of constant absolute value is constant (For instance, show that if $f(z)$ is analytic in a domain D and $|f(z)| = k = \text{constant}$ in D , then $f(z)$ is constant in D .)

Sol:

By assumption,

$$|f|^2 = |u+iv|^2 = u^2 + v^2 = k^2. \quad \text{By diff}$$

$$2uu_x + 2vv_x = 0 \Rightarrow \boxed{uu_x + vv_x = 0} \rightarrow \textcircled{1}$$

$$2uu_y + 2vv_y = 0 \Rightarrow \boxed{uu_y + vv_y = 0} \rightarrow \textcircled{2}$$

By using C.R Eqs.

$$u_x = v_y, \quad u_y = -v_x. \rightarrow \textcircled{A}$$

from $\textcircled{1}$ and $\textcircled{2}$

$$uu_x + v(-u_y) = 0 \rightarrow \textcircled{3}$$

$$uu_y + v(u_x) = 0 \rightarrow \textcircled{4}$$

To get rid of u_x Multiply $\textcircled{3}$ by v and

Multiply $\textcircled{4}$ by u .

$$\begin{array}{r} vxu_x - v^2u_y = 0 \\ + u^2u_y + uvu_x = 0 \\ \hline - (v^2 + u^2)u_y = 0 \end{array}$$

$$\Rightarrow \boxed{(u^2 + v^2)u_y = 0} \rightarrow \textcircled{*}$$

To get rid of uy Multiply eq (3) by u
and eq (4) by v and then add.

$$\begin{aligned} u^2 u_x + v u (-u_y) &= 0 \\ v u u_y + v^2 u_x &= 0 \\ \hline (u^2 + v^2) u_x &= 0 \rightarrow \text{**} \end{aligned}$$

Again from
Eqs (1), (2) and (A)

$$\begin{aligned} u(v_y) + v v_x &= 0 \rightarrow (5) \\ u(-v_x) + u v_y &= 0 \rightarrow (6) \end{aligned}$$

After simplifications.

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$$(u^2 + v^2) v_y = 0 \rightarrow \text{***} \quad \text{and} \quad (v^2 + u^2) v_x = 0 \rightarrow \text{****}$$

If $k^2 = u^2 + v^2 = 0$ then $u = v = 0$

$$\text{hence } f = 0$$

If $k^2 = u^2 + v^2 \neq 0$ then.

$$u_x = u_y = v_x = v_y = 0 \rightarrow (7)$$

Together

$$\frac{\partial u}{\partial x} = 0$$

$$u = f(y)$$

$$\frac{\partial u}{\partial y} = f'(y)$$

given

$$0 = f'(y) \Rightarrow f(y) = C$$

So

$$\begin{aligned} u &= C \\ \text{Similarly} \\ v &= C \\ \text{Hence} \\ f &\text{ is constant} \end{aligned}$$

We mention that, if we use the polar form $z = r(\cos \theta + i \sin \theta)$ and set $f(z) = u(r, \theta) + iv(r, \theta)$, then the **Cauchy–Riemann equations** are (Prob. 1)

$$(7) \quad \begin{aligned} u_r &= \frac{1}{r} v_\theta, \\ v_r &= -\frac{1}{r} u_\theta \end{aligned} \quad (r > 0).$$

Laplace's Equation. Harmonic Functions

The great importance of complex analysis in engineering mathematics results mainly from the fact that both the real part and the imaginary part of an analytic function satisfy Laplace's equation, the most important PDE of physics. It occurs in gravitation, electrostatics, fluid flow, heat conduction, and other applications (see Chaps. 12 and 18).

THEOREM 3

Laplace's Equation

If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then both u and v satisfy Laplace's equation

$$(8) \quad \nabla^2 u = u_{xx} + u_{yy} = 0$$

(∇^2 read “nabla squared”) and

$$(9) \quad \nabla^2 v = v_{xx} + v_{yy} = 0,$$

in D and have continuous second partial derivatives in D .

PROOF Differentiating $u_x = v_y$ with respect to x and $u_y = -v_x$ with respect to y , we have

$$(10) \quad u_{xx} = v_{yx}, \quad u_{yy} = -v_{xy}.$$

Now the derivative of an analytic function is itself analytic, as we shall prove later (in Sec. 14.4). This implies that u and v have continuous partial derivatives of all orders; in particular, the mixed second derivatives are equal: $v_{yx} = v_{xy}$. By adding (10) we thus obtain (8). Similarly, (9) is obtained by differentiating $u_x = v_y$ with respect to y and $u_y = -v_x$ with respect to x and subtracting, using $u_{xy} = u_{yx}$. ■

Solutions of Laplace's equation having *continuous* second-order partial derivatives are called **harmonic functions** and their theory is called **potential theory** (see also Sec. 12.11). Hence the real and imaginary parts of an analytic function are harmonic functions.

If two harmonic functions u and v satisfy the Cauchy–Riemann equations in a domain D , they are the real and imaginary parts of an analytic function f in D . Then v is said to be a **harmonic conjugate function** of u in D . (Of course, this has absolutely nothing to do with the use of “conjugate” for $\bar{z}$.)

Example: 4

How to find a Harmonic Conjugate Function by the Cauchy-Riemann Equations.

Verify that $u = x^2 - y^2 - y$ is Harmonic in the complex plane and find a harmonic conjugate function v of u .

Sol:

To find Harmonic we have to satisfy

$$\nabla^2 u = u_{xx} + u_{yy} = 0 \quad \text{--- (1)}$$

$$u = x^2 - y^2 - y$$

$$u_x = 2x, \quad u_y = -2y - 1$$

$$u_{xx} = 2, \quad u_{yy} = -2$$

from (1) $2 - 2 = 0$ satisfied.

To find Harmonic conjugate function v

we will use C.R Eqs

$$u_x = v_y \quad \text{--- (2)}, \quad u_y = -v_x \quad \text{--- (3)}$$

Consider eq (2)

$$\begin{array}{c} u_x = v_y \\ \downarrow \\ \boxed{v_y = 2x} \end{array}$$

$$\text{As } \frac{\partial v}{\partial y} = 2x \quad \text{--- (4)}$$

Integrate eq (4) w.r.t. y.

$$\int \frac{\partial v}{\partial y} dy = \int 2x dy + g(x)$$

$$v = 2xy + g(x) \rightarrow (5)$$

To find the value of $g(x)$ diff w.r.t. of eq (5).

$$\frac{\partial v}{\partial x} = 2y + g'(x) \rightarrow (6)$$

By using eq (3)

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$-\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

$$-(2y-1) = \frac{\partial v}{\partial x}$$

from (6)

$$2y+1 = 2y + g'(x)$$

$$g'(x) = 1$$

$$\boxed{g(x) = x + c}$$

from (5)

$$\boxed{v = 2xy + x + c} \rightarrow \text{harmonic conjugate ftn of } u.$$

and the corresponding Analytic ftn.

$$f(z) = u + iv$$

$$= (x^2 - y^2 - y) + i(2xy + x + c)$$

Recommended Exercises

13.3 and 13.4

Book: advanced engineering Mathematics